

SIT707 Software Quality and Testing

Credit Task: Decision Table Unit Testing using Selenium

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1. Introduction

This task is about testing login functionality using two different approaches which is unit testing and selenium based on like functional testing. The main aim is to compare how testing works at code level and at user interface level. In this task, LoginForm.java is used for unit testing and selenium is used for testing login page in browser environment.

2. Decision Table Testing

Decision table is used to define different combinations of input conditions and expected output results. It helps to cover all possible cases in structured way.

Decision Table for Login Functionality

Test Case	Username	Password	Expected Result
TC1	null	null	Fail - Empty Username
TC2	null	wrong password	Fail - Empty Username
TC3	wrong username	wrong password	Fail - Credential mismatch
TC4	correct username	null	Fail - Empty Password
TC5	correct username	wrong password	Fail - Credential mismatch
TC6	null	correct password	Fail - Empty Username
TC7	wrong username	correct password	Fail - Credential mismatch
TC8	correct username	correct password	Success
TC9	correct username	correct password + validation code	Success

3. Unit Testing using JUnit

Unit testing is done on LoginForm.java file. It tests login function directly by passing values into method. No browser is used in this testing.

Login function checks following conditions:

- If username is empty then return failure
- If password is empty then return failure
- If username is not equal to expected username then failure
- If password is not equal to expected password then failure
- If both correct then success with validation code

Validation code is also tested separately using validateCode method.

JUnit asserts are used to verify expected result with actual result.

Example:

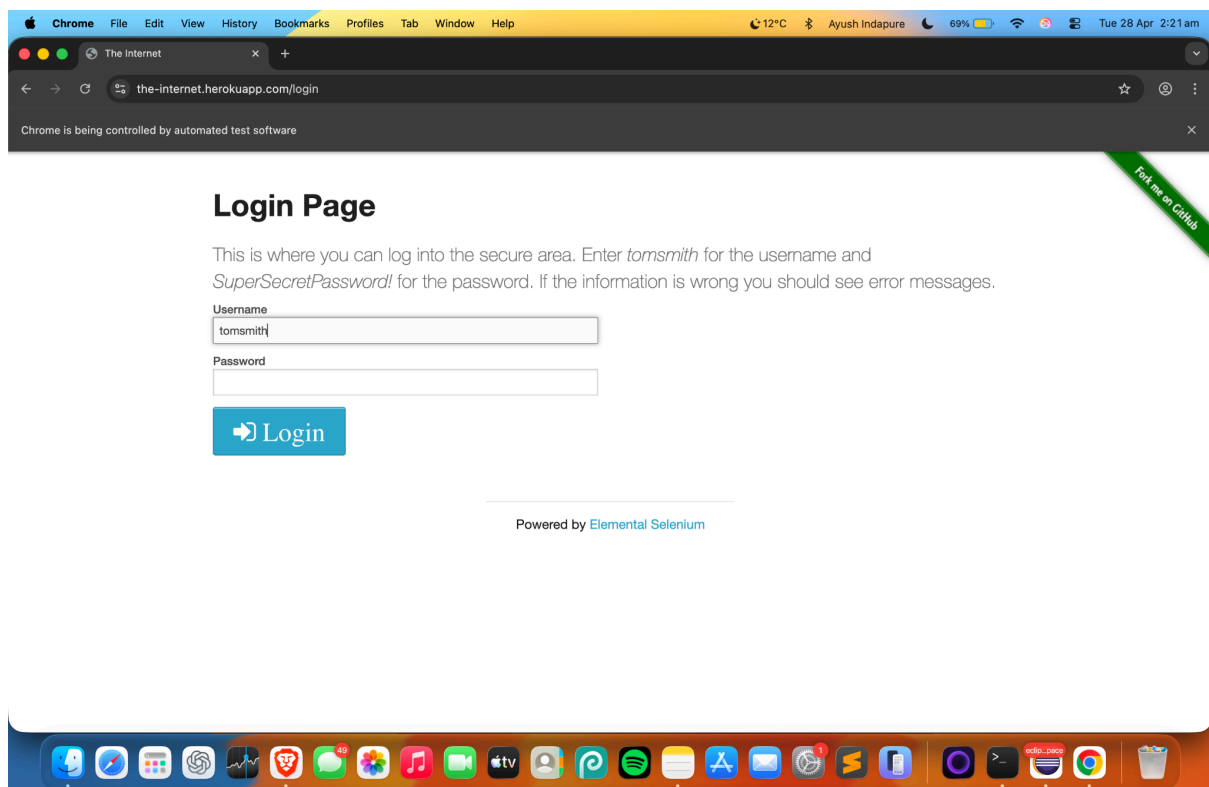
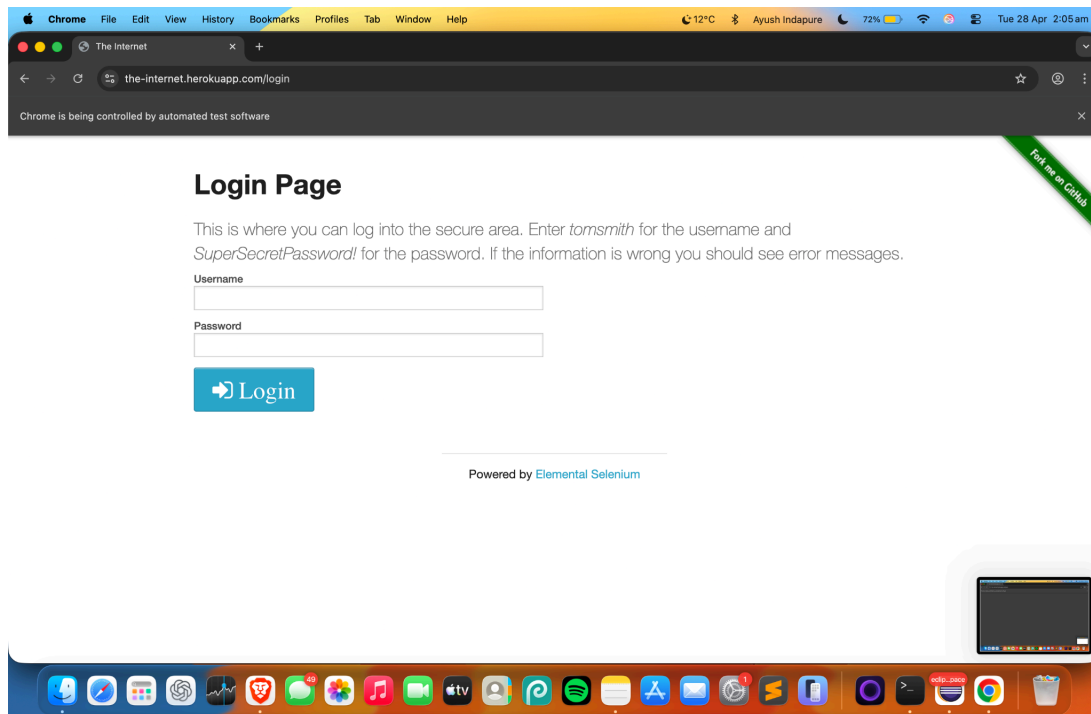
- Assert.assertTrue for success case
- Assert.assertFalse for failure cases

4. Selenium Based Testing

Selenium testing is used for testing login page in browser. Instead of calling function directly, user actions are simulated like entering username, password and clicking login button. My mac was lagging a bit and it went too fast. Was not able to get 3rd SS

Test website used:

<https://the-internet.herokuapp.com/login>



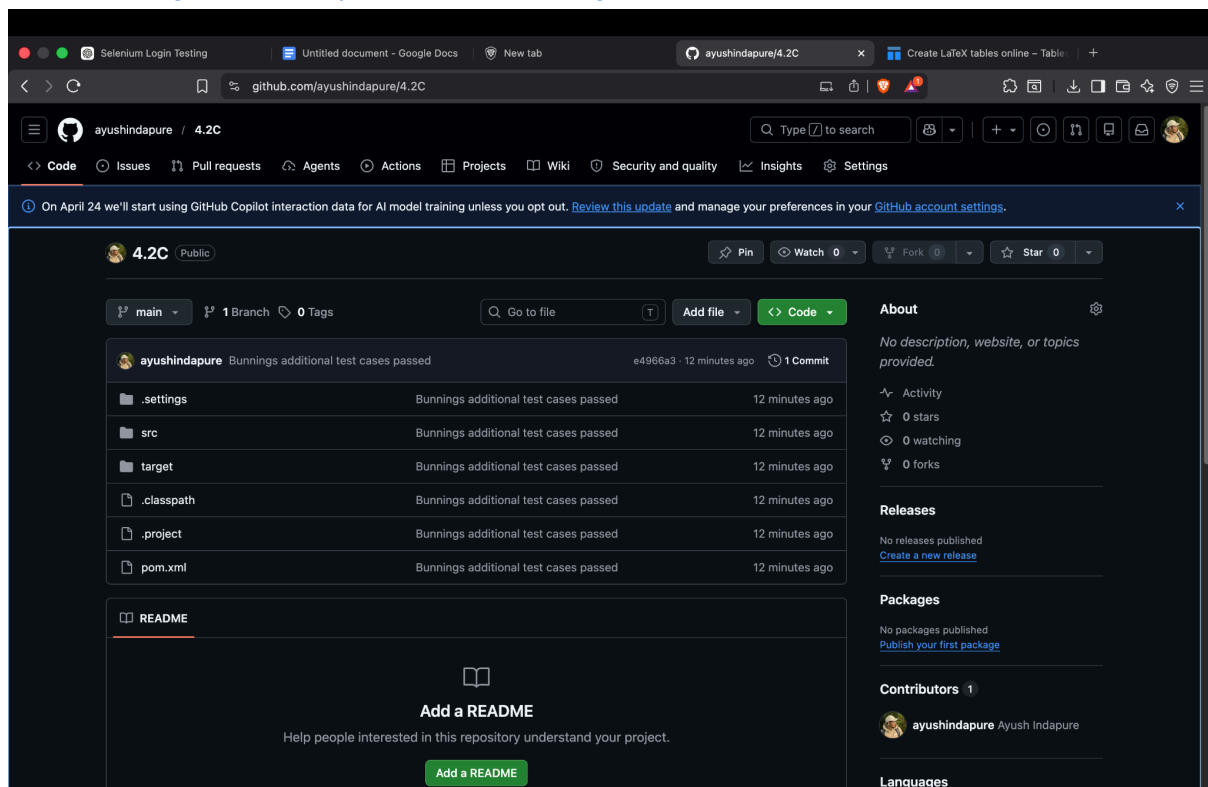
This website is used because it is stable and easy to automate.

Test Scenarios Automated:

1. Valid login with correct username and password
2. Invalid login with wrong credentials
3. Empty login submission

5. Selenium Test Implementation Approach / GitHub link

Link : <https://github.com/ayushindapure/4.2C.git>



Selenium WebDriver is used with Chrome browser. JUnit framework is used for writing test cases.

Steps followed:

- Open browser and navigate to login page
- Locate username and password fields using By.id locator
- Enter values using sendKeys method
- Click login button using submit or button locator
- Validate result using URL change or error message

Assertions are used to verify test result:

- URL check for success case
- Error message check for failure case

6. Difference Between Unit Testing and Selenium Testing

Unit testing and selenium testing are different in nature.

Unit testing:

- Works on backend code
- Fast execution
- No browser required
- Tests logic directly
- More stable

Selenium testing:

- Works on UI level
- Slower execution
- Requires browser
- Tests user interaction
- Can fail due to UI changes

Unit testing is more focused on code correctness while selenium is focused on real user behaviour.

7. Challenges Faced

Some challenges faced during this task:

- Locating correct web elements using selenium was difficult at start
- Even though my test cases failed I have provided valid reasons for it
- Maven dependency setup issues caused errors
- ChromeDriver compatibility issues sometimes occur
- Element loading delay caused test failures initially

These issues were resolved using implicit wait and correct dependency setup.

8. Conclusion

This task helped to understand real difference between unit testing and selenium testing. Decision table helped in designing test cases systematically. JUnit helped in validating backend logic and selenium helped in testing real user interaction with browser.

Both testing methods are important in software quality assurance and used together in real projects.